

Companion LIII: E–B Mode Conversion in the Relaxation-Driven Cyclic Cosmology

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Abstract

Weak gravitational lensing converts CMB E-modes into B-modes, providing a powerful probe of the gravitational potential along the line of sight. In the standard cosmological model, this conversion is determined by the lensing potential and the geometry of the deflection field. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential modifies the lensing kernel, the deflection field, and the E–B mode coupling in a characteristic way. This Companion develops a continuous description of E–B mode conversion in RDCC, deriving how the relaxation scale influences the lensing-induced B-mode spectrum, the mode-coupling structure, and the coherence of the deflection field. The resulting framework provides a distinctive observational signature of the RDCC gravitational field.

1 Introduction

CMB polarization provides a unique window into the geometry and dynamics of the Universe. E-modes arise from scalar perturbations, while B-modes are generated by tensor perturbations or by the lensing-induced conversion of E-modes into B-modes. The latter process is sensitive to the gravitational potential along the line of sight and provides a direct probe of large-scale structure.

In the standard cosmological model, the E–B mode conversion is determined by the lensing potential and the geometry of the deflection field. In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the decay of small-scale modes and enhances the coherence of large-scale modes. This modifies the lensing potential, the deflection field, and the E–B mode coupling in a scale-dependent manner, producing a distinctive signature in the lensing-induced B-mode spectrum.

2 The RDCC Lensing Potential and Deflection Field

The lensing potential is given by

$$\phi(\hat{\mathbf{n}}) = -2 \int_0^{\chi_*} d\chi \frac{\chi_* - \chi}{\chi_* \chi} [\Phi + \Psi](\chi, \hat{\mathbf{n}}). \quad (1)$$

In RDCC, the metric potentials evolve as

$$\Phi_{\text{RDCC}}(k, a) = \Phi(k, a) \left[1 - \chi_\Phi \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right], \quad (2)$$

$$\Psi_{\text{RDCC}}(k, a) = \Psi(k, a) \left[1 - \chi_\Psi \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (3)$$

The deflection field becomes

$$\mathbf{d}_{\text{RDCC}} = \nabla \phi_{\text{RDCC}} = \mathbf{d} \left[1 - \chi_d \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (4)$$

3 E–B Mode Conversion in RDCC

Lensing converts E-modes into B-modes according to

$$B_{\ell m} = \sum_{\ell' m'} \sum_{LM} \phi_{LM} \mathcal{G}_{\ell \ell' L}^{mm' M} E_{\ell' m'}, \quad (5)$$

where $\mathcal{G}$ is a geometric coupling coefficient.

In RDCC, the modified lensing potential produces

$$B_{\ell m}^{\text{RDCC}} = B_{\ell m} \left[1 - \chi_B \left(\frac{\ell}{\ell_{\text{rel}}} \right)^\alpha \right]. \quad (6)$$

This modification suppresses small-scale B-modes and enhances large-scale coherence.

4 The RDCC B-Mode Power Spectrum

The lensing-induced B-mode power spectrum is given by

$$C_\ell^{BB} = \int d^2 \ell' [\ell' \cdot (\ell - \ell')]^2 C_{\ell'}^{EE} C_{|\ell - \ell'|}^{\phi\phi}. \quad (7)$$

In RDCC, the lensing potential power spectrum becomes

$$C_L^{\phi\phi, \text{RDCC}} = C_L^{\phi\phi} \left[1 - \chi_{\phi\phi} \left(\frac{L}{L_{\text{rel}}} \right)^\alpha \right]. \quad (8)$$

The B-mode spectrum becomes

$$C_\ell^{BB, \text{RDCC}} = C_\ell^{BB} \left[1 - \chi_{BB} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^\alpha \right]. \quad (9)$$

5 Large-Scale Coherence and RDCC Signatures

The relaxation mechanism enhances the coherence of large-scale modes. The lensing-induced B-mode spectrum therefore exhibits:

- enhanced power at large angular scales,
- suppressed power at small scales,
- a characteristic turnover near ℓ_{rel} ,
- a modified phase structure in E–B coupling.

These signatures provide a direct observational probe of the RDCC gravitational field.

6 Conclusion

The E–B mode conversion in the Relaxation-Driven Cyclic Cosmology reflects the scale-dependent evolution of the gravitational potential. The relaxation mechanism modifies the lensing potential, the deflection field, and the E–B mode coupling in a scale-dependent manner, producing distinctive signatures in the lensing-induced B-mode spectrum. These effects provide a direct observational window into the relaxation scale and the temporal structure of the RDCC gravitational field. The present Companion establishes the theoretical foundation for future studies of CMB polarization, lensing reconstruction, and the interplay between matter and gravity in RDCC.

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